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Register Number: 1 2 0 0 7

Class : 12

COMMON HALFYEARLY EXAMINATION 2022 - 23

CHEMISTRY

PART – I

Time Allowed : 3.00 Hours] [Max. Marks : 70

15x1=15

I. **Choose the correct answer.**

1. In the extraction of aluminium from alumina by electrolysis, cryolite is added to
   a) Lower the melting point of alumina b) Remove impurities from alumina
   c) decrease the electrical conductivity d) Increase the rate of reduction.
2. The isotope - is used as moderator in Nuclear reactors.
   a) $_5B^{10}$ b) $_6C^{11}$ c) $_{13}Al^{27}$ d) $_{14}Si^{28}$
3. P_4O_6 reacts with cold water to give
   (a) H_3PO_3 (b) $H_4P_2O_7$ (c) HPO_3 (d) H_3PO_4
4. Equivalelnt weight of $KMnO_4$ in basic medium
   a) 31.6 b) 158 c) 52.67 d) 79
5. IUPAC name of the complex $K_3[Al(C_2O_4)_3]$ is
   a) Potassium trioxalato aluminium (III) b) Potassium trioxalato aluminate (II)
   c) Potassium trisoxalato aluminate (III) d) Potassium trioxalato anuminate (III)
6. The vacant space in bcc Lattice unit cell is
   a) 48% b) 23% c) 32% d) 26%
7. What is the activation energy for a reaction if its rate doubles when the temperature is raised from 200 K to 400K? (R = 8.314 JK^{-1} mol^{-1})
   a) 234.65 K J mol^{-1} b) 434.65 K J mol^{-1} c) 2.305 K J mol^{-1} d) 334.65 K J mol^{-1}
8. Dissociation constant of NH_4OH is 1.8×10^{-5} the hydrolysis constant of NH_4Cl would be
   a) 1.8×10^{19} b) 5.55×10^{-10} c) 5.55×10^{-5} d) 1.80×10^{-5}
9. The electrode which acts as anode in lithium - ion battery is --------
   a) Nickel b) Zine c) Porus graphite d) Copper rod
10. **Match the following:**

A)	V_2O_5	(i)	High density poluethylene
B)	Ziegler - Nalta	(ii)	PAN
C)	Peroxide	(iii)	NH_3
D)	Finely divided Fe	(iv)	H_2SO_4

	A	B	C	D
a)	(iv)	(i)	(ii)	(iii)
b)	(i)	(ii)	(iv)	(iii)
c)	(ii)	(iii)	(iv)	(i)
d)	(iii)	(iv)	(ii)	(i)

11. Williamson synthesis of preparing dimethyl ether is a / an /
    a) SN^1 reactions b) SN^2 reaction c) electrophilic addition d) electrophilic substitution
12. Which one of the following reaction is an example of disproportionation reaction
    a) Aldol condensation b) Cannizaro reaction
    c) Benzoin condensation d) none of these.
13. Which one of the following is most basic?
    a) 2, 4 - dichloro aniline b) 2, 4 - dimethyl aniline
    c) 2, 4 - dinitro aniline d) 2, 4 - dibromo aniline
14. The chemical name of vitamin B_8
    a) Retinol b) Thiamine c) Pantothanic acid d) Biotin
15. Regarding cross - linked or network polymers, which of the following statement is incorrect?
    a) Examples are bakelite and melamine
    b) They are formed from bi and tri functional monomers
    c) They contain covalent bonds between various linear polymer chains.
    d) They contain strong covalent bonds in their polymer chains.

Kindly send me your questions and answerkeys to us : Padasalai.net@gmail.com

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PART - II

II. Answer any six questions. Question No. 24 is compulsory. 6x2=12

16. Give the limitation of Ellingham diagram.
17. What is insert pair effect?
18. What is crystal field splitting energy?
19. Differentiate crystalline solids and amorphons solids?
20. Explain Helmholtz double layer.
21. Explain Kolbe's electrolytic reaction.
22. Write a short note on peptide bond.
23. Write a note on vulcanization of rubber.
24. Calculate the electro chemical eqiuvalent of silver in silver nitrate.

PART - III

III Answer any six questions. Question No. 33 is compulsory. 6x3=18

25. GIve the uses of Silicones.
26. How will you identify borate radical?
27. How is pure phosphine prepared from phosphorous acid?
28. Write the differences between order and molecularity of reaction.
29. Discuss the Lowry - Bronsted concept of acids and bases.
30. Explain the factors affecting electrolytic conductance?
31. Explain Tautomevism.
32. How is Teflon prepared?
33. Identify A, B and C Ettanoic acid $\xrightarrow{SoCl_2}$ A $\xrightarrow{Pd / BaSO_4}$ B $\xrightarrow{NaOH}$ C

PART - IV

IV Answer all the questions. 5x5=25

34. (a) i) Describe a method for refining Nickel.
    ii) What is Auto reduction?

    OR

    (b) i) Complete the following reactions.

    A) $H_2B_4O_7 \xrightarrow{\text{Red hot}}$

    B) $B(OH)_3 + NH_3 \longrightarrow$

    ii) Write about Holmes signal.

35. (a) (i) Compare Lanthanides and Actinides.
    (ii) Explain the structure of permaganate ion.

    OR

    (b) Write the postulates of valence bond theory.

36. (a) (i) Distinguish between tetrehedral and octahedral voids.
    (ii) Explain Frenkel deffect.

    OR

    (b) Derive Integrated rate law for the first order reaction.

37. (a) (i) Derive an expression for ostwalds's dilution law.
    (ii) Is it possible to store copper sulphate in an iron vessel for a long time?
    Given : $E^o_{Cu^{2+}/Cu}$ = 0.34 V and $E^o_{Fe^{2+}/Fe}$ = - 0.44 V

    OR

    (b) What are the characteristics of a catalyst.

38. (a) Explain the mechanism of cannizaro reaction.

    OR

    (b) i) Write the reduction products are obtained when nitrobenzene is reduced in acid medium and neutral medium
    ii) Write short notes on Gomberg reaction.

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